

1.

unsalted : 5 mg Na / oz

$$\frac{5 \text{ mg Na}}{1 \text{ oz}} \cdot \frac{1 \text{ g}}{1000 \text{ mg}} \cdot \frac{10 \text{ oz}}{0.0625 \text{ lbm}} = 0.08 \text{ g / lbm}$$

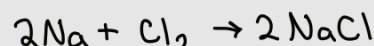
Lightly Salted : 75 mg Na / oz

Salted : 150 mg Na / oz

$$\left\{ \begin{array}{l} 3/4 \text{ tsp NaCl} \\ 1 \text{ lbm peanuts unsalted} \end{array} \right.$$

$$1 \text{ tsp NaCl} = 6 \text{ g}$$

$$\frac{3}{4} \text{ tsp NaCl} \cdot \frac{6 \text{ g}}{1 \text{ tsp NaCl}} = \frac{9}{2} = 4.5 \text{ g}$$



a)

$$\frac{75 \text{ mg Na}}{1 \text{ oz}} \cdot \frac{1 \text{ g}}{1000 \text{ mg}} \cdot \frac{10 \text{ oz}}{0.0625 \text{ lbm}} = 1.2 \text{ g Na / lbm}$$

$$\frac{1.2 \text{ g Na}}{1 \text{ lbm}} \cdot \frac{1 \text{ mol Na}}{22.99 \text{ g Na}} \cdot \frac{2 \text{ mol NaCl}}{2 \text{ mol Na}} \cdot \frac{58.44 \text{ g NaCl}}{1 \text{ mol NaCl}} = 3.05 \text{ g NaCl / lbm}$$

$$\frac{3.05 \text{ g NaCl}}{1 \text{ lbm}} \cdot 1 \text{ lbm} = 3.05 \text{ g NaCl}$$

$$(4.5 + 0.08) - 3.05 = 1.45 \text{ g NaCl}$$

- 3 pts

$$1.45 \text{ g NaCl} \cdot \frac{1 \text{ tsp}}{6 \text{ g NaCl}} = 0.242 \text{ tsp NaCl needed}$$

b)

$$\frac{150 \text{ mg Na}}{1 \text{ oz}} \cdot \frac{1 \text{ g Na}}{1000 \text{ mg Na}} \cdot \frac{10 \text{ oz}}{0.0625 \text{ lbm}} = 2.4 \text{ g Na / lbm}$$

$$\frac{2.4 \text{ g Na}}{1 \text{ lbm}} \cdot \frac{1 \text{ mol Na}}{22.99 \text{ g Na}} \cdot \frac{2 \text{ mol NaCl}}{2 \text{ mol Na}} \cdot \frac{58.44 \text{ g NaCl}}{1 \text{ mol NaCl}} = 6.1 \text{ g NaCl / lbm}$$

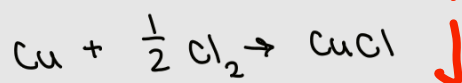
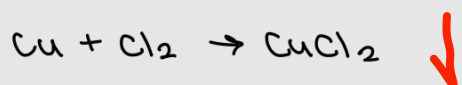
$$\frac{6.1 \text{ g NaCl}}{1 \text{ lbm}} \cdot 1 \text{ lbm} = 6.1 \text{ g NaCl}$$

No more table salt needed
as it has overshoot the
original amount

no part c, - 6 pts

P1: 11/20

2)



$$x_{\text{CuCl}_2} = \frac{168.047}{200} = 0.84$$

$$x_{\text{CuCl}} = \frac{31.953}{200} = 0.16$$

$$\frac{x_{\text{CuCl}_2}}{x_{\text{CuCl}}} = \frac{0.84}{0.16} = 5.25$$

$$5.25 : 1 \quad \checkmark$$

$$m_{\text{Cu}_2\text{Cl}} + m_{\text{CuCl}} = 200 \quad \checkmark$$

$$x_1 = \frac{m_{\text{Cu}}}{m_{\text{CuCl}_2}} = \frac{63.55}{134.45} = 0.473 \quad \checkmark$$

$$x_2 = \frac{m_{\text{Cu}}}{m_{\text{CuCl}}} = \frac{63.55}{99} = 0.642 \quad \checkmark$$

$$0.473 m_{\text{CuCl}_2} + 0.642 m_{\text{CuCl}} = 100 \quad \checkmark$$

$$m_{\text{CuCl}_2} = 168.047$$

$$m_{\text{CuCl}} = 31.953$$

P2: 20/20

3)

a)

$$\Delta H = \int_{T_1}^{T_2} (A + BT + CT^2 + DT^3) dT \quad \checkmark$$

$$= \left(AT + \frac{1}{2} BT^2 + \frac{1}{3} CT^3 + \frac{1}{4} DT^4 \right) \Big|_{300}^{700}$$

$$= (21,033.1) - (8,568.41)$$

$$\Delta H = 12,464.7 \text{ J/mol} \quad \checkmark$$

$$A = 28.11$$

$$B = -3.68 \times 10^{-6}$$

$$C = 1.746 \times 10^{-5}$$

$$D = -1.065 \times 10^{-8}$$

b)

$$t = \frac{T}{1000}$$

$$T = 1000t$$

$$c_p(t) = A + B(1000t) + C(1,000,000t^2) + D(1,000,000,000t^3) \quad \checkmark$$

$$\Delta H = \int_{300}^{700} c_p(t) dt = \int_{300}^{700} (A + B(1000t) + C(1,000,000t^2) + D(1,000,000,000t^3)) dt$$

$$\Delta H = \left(At + \frac{1000}{2} Bt^2 + \frac{1,000,000}{3} Ct^3 + \frac{1,000,000,000}{4} Dt^4 \right) \Big|_{300}^{700}$$

$$\Delta H = (21,033.1) - (8,568.41)$$

$$\Delta H = 12,464.7 \text{ J/mol} \quad \checkmark$$

P3: 20/20

4)

$$\int_1^h \frac{1}{40 - 4\sqrt{x-1}} dx = -\frac{\sqrt{h-1}}{2} - 5 \log(10 - \sqrt{h-1}) + 5 \log(10)$$

$$u = \sqrt{x-1} \quad u(1) = 0$$

$$u^2 = x-1 \quad u(h) = \sqrt{h-1}$$

$$2u \, du = dx$$

$$= \int_0^{\sqrt{h-1}} \frac{2u}{40-4u} \, du$$

$$= \int_0^{\sqrt{h-1}} \frac{2u}{4(10-u)} \, du$$

$$= \int_0^{\sqrt{h-1}} \frac{u}{2(10-u)} \, du$$

$$= \frac{1}{2} \int_0^{\sqrt{h-1}} \frac{u}{10-u} \, du$$

$$= \frac{1}{2} \int_0^{\sqrt{h-1}} -1 + \frac{10}{10-u} \, du$$

$$= \frac{1}{2} \left(\int_0^{\sqrt{h-1}} -1 \, du + \int_0^{\sqrt{h-1}} \frac{10}{10-u} \, du \right)$$

$$= \frac{1}{2} \left((-u) \Big|_0^{\sqrt{h-1}} + \int_0^{\sqrt{h-1}} \frac{10}{w} \, du \right)$$

$$= \frac{1}{2} \left(-\sqrt{h-1} + (10 \ln(10-u)) \Big|_0^{\sqrt{h-1}} \right)$$

$$= \frac{1}{2} \left(-\sqrt{h-1} + 10 \ln(10-\sqrt{h-1}) - 10 \ln(10) \right)$$

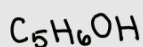
$$= -\frac{1}{2} \sqrt{h-1} + 5 \ln(10-\sqrt{h-1}) - 5 \ln(10)$$

$$w = 10-u$$

$$dw = -du$$

P4: 20/20

5)



$$7 \text{ Kcal/g}$$

$$\Delta H_g = -1,235 \text{ KJ/mol}$$

$$\Delta H_l = -1,367 \text{ KJ/mol}$$

$$T = 25^\circ\text{C}$$

$$P = 1 \text{ atm}$$

$$\text{molar mass of ethanol} = 46.07 \text{ g/mol}$$

$$\text{Kcal/g to KJ/mol}$$

$$\Delta H = \frac{7 \text{ Kcal}}{1 \text{ g}} \cdot \frac{4.184 \text{ KJ}}{1 \text{ Kcal}} \cdot \frac{46.07 \text{ g}}{1 \text{ mol}} = 1,349.3 \frac{\text{KJ}}{\text{mol}}$$

I can conclude this mixture of ethanol is likely to be a liquid state since the magnitude of ΔH_l and ΔH are closer in value than ΔH_g

P5: 20/20